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Particle swarm optimization algorithm for design of an adaptive Kanban system based on optimization via simulation

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ABSTRACT

Manufacturing systems face challenges in adapting to fluctuating demand, making traditional fixed-parameter Kanban systems inefficient. This study addresses these limitations by proposing an adaptive Kanban-based pull control mechanism that dynamically adjusts the number of Kanbans based on inventory levels and backorders. To design an adaptive Kanban system, a metaheuristic optimization approach using particle swarm optimization is integrated with simulation to account for stochastic and time-varying customer demand. Previous studies have either overlooked demand fluctuations in Kanban systems or underutilized the advantages of particle swarm optimization. The results demonstrate that the proposed algorithm significantly reduces total production costs compared to existing metaheuristic methods. These findings underscore the importance of integrating simulation and optimization to enhance decision-making in manufacturing, providing a robust framework for improving production flow control in uncertain environments.

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1. Introduction

In the context of the 4th industrial revolution, several countries have allocated significant investments to address major technical and scientific challenges with the goal of improving the performance of their companies [1]. Despite the opportunities presented by this revolution, companies must confront significant difficulties caused by market evolution, production uncertainties, and unpredictable demand fluctuation. To remain efficient and competitive, production systems must be capable of adapting to face these uncertainties.

This adaptability can be achieved through the implementation of adaptive systems that leverage optimization techniques via simulation. The use of simulation to optimize dynamic models offers several significant advantages, particularly in the management of complex or dynamic systems. Simulation allows a detailed examination of system behavior under different scenarios, incorporating uncertainties and dynamics that may be neglected by traditional models [2,3]. By combining optimization techniques with simulation, decision-makers can identify optimal or near-optimal solutions while assessing their feasibility and robustness through simulation experiments. This feedback loop enhances the decision-making process by allowing adjustments to be made based on performance metrics derived from simulation results. Furthermore, optimization coupled to simulation can effectively achieve conflicting objectives, such as cost reduction and customer satisfaction [4]. Overall, simulation is an efficient and powerful tool for improving responsiveness and performance in dynamic environments [5–8].

The Kanban system is widely recognized as an effective tool for managing production processes, particularly within the framework of Just-In-Time (JIT) manufacturing [9]. By facilitating a pull-based control mechanism centered on producing products precisely when customers need them, primarily by eliminating sources of waste [10], Kanban helps organizations minimize Work in Progress (WIP) and streamline operations to meet customer demands efficiently [11]. However, while Kanban has proven beneficial in stable environments with predictable demand [12], it presents limitations in addressing the complexities of dynamic environments conditions. As customer preferences changes and demand becomes increasingly stochastic, traditional Kanban systems can lead to inefficiencies such as suboptimal inventory levels and resource underutilization. To overcome these limitations, previous studies aim to solve the rigidity of traditional approaches, providing a more responsive and cost-effective solution.

Despite numerous studies in production system optimization, the practical implementation of approaches that effectively address dynamic and stochastic environments is still a challenge for manufacturing management. Many existing studies have highlighted the limitations of traditional systems in dealing with such conditions, particularly when attempting to balance conflicting objectives such as minimizing costs while maximizing customer

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satisfaction. Identifying optimal parameters in adaptive production systems is a complex challenge, especially under fluctuating and uncertain demand conditions. Although optimization via simulation has proven to be a powerful technique for addressing these challenges by coupling system dynamics with advanced optimization methods, it has yet to see widespread adoption in practice.

This study aims to fill this gap by leveraging the Particle Swarm Optimization (PSO) algorithm to optimize an adaptive Kanban system. PSO, inspired by the collective behavior of swarms, is an efficient and robust method for solving complex, non-linear problems. Each particle in the algorithm represents a potential solution, and through iterative interactions, the swarm collectively searches for the best solution based on a fitness function [13]. In the context of adaptive Kanban systems, PSO shows great potential for optimizing decision parameters, enabling systems to dynamically adjust and improve performance under varying demand conditions.

To achieve these objectives, this study employs optimization via simulation, a method particularly suited to capturing the complexities of stochastic and time-varying demand. Simulation provides a detailed representation of system behavior under different scenarios, incorporating uncertainties and evaluating the robustness and feasibility of solutions. By integrating PSO with simulation, this research not only demonstrates the effectiveness of this approach but also offers a practical framework for enhancing the adaptability and performance of production systems. In order to evaluate the performance of the proposed approach, a comparative study based on the objective function was conducted. The PSO algorithm was compared to other algorithms predeveloped in the existing literature.

The structure of the study is as follows. Section 2 provides an analysis of the current literature on adaptive card-based pull control systems, with a focus on approaches for dynamically adjusting the number of cards in the absence of production forecasts. In Section 3, we outline our methodology, which combines an optimization via simulation approach with a metaheuristic approach. The Particle Swarm Optimization algorithm is elaborated. In Section 4, we apply our approach to examples of adaptive Kanban systems drawn from the literature in order to test the performance of the proposed approach. The simulation results are analyzed and discussed in section 5. A comparison of the proposed approach with other methods developed in the literature demonstrates the superiority of our approach. Finally, the study concludes with a summary of the results and a discussion of potential new future works.

2. Literature review

2.1. Problem statement and related research

The traditional Kanban system (TKS) is a pull-based manufacturing system that uses a set of cards to signal the need for more materials when a production stage is complete. It effectively control the flow of materials in a production process, allowing for the efficient use of resources and reducing waste. However, the traditional Kanban system with fixed number of card has limitations when it comes to dealing with uncertain and fluctuation demand. To address these challenges, several studies have proposed card controlling approaches. In these methods, the number of Kanban cards in circulation is adjusted to accommodate the dynamic and unpredictable changes in the production environment. Such systems are often categorized as flexible pull control systems [14,15], reactive systems [16], or adaptive systems [17–20] offering different strategies to improve responsiveness and adaptability in managing production flows under varying conditions.

Adaptive card production control policies have been widely discussed in the literature to dynamically adjust the number of cards in response to environmental changes. The literature provides various approaches and information on the implementation and effectiveness of these adaptive systems.

Tardif and Maaseidvaag [17] designed the Adaptive Kanban System (AKS) to manage variable supply and demand conditions. The proposed approach aims to responds to changes in demand by dynamically adjusting the number of Kanban based on inventory and backorder levels. Figure 1 shows the AKS model. The number of cards used is allowed to vary based on the current stock level and pending orders. When a demand arrives at time t to the manufacturing system, the inventory level N(t) is calculated. N(t) represent the quantity of finished pieces in excess of pending orders at time t expressed by (P(t) – D(t)).

- If the inventory level N(t) is below the release threshold (R), $N(t) \le R$ and an extra card is available $x(t) > 0$, then an extra card is released and sent to MP. The demand is satisfied as soon as a finished product is available
- If the inventory level N(t) exceeds the capture threshold (C), $N(t) \ge C$ and an extra card is available $x(t) < E$, then the demand is satisfied as soon as a finished product is available, the card attached to the product is captured and queue E is updated.

The queue P contains finished pieces, and queue D contains pending orders. The single-stage adaptive mechanism uses K initial cards and E additional cards.

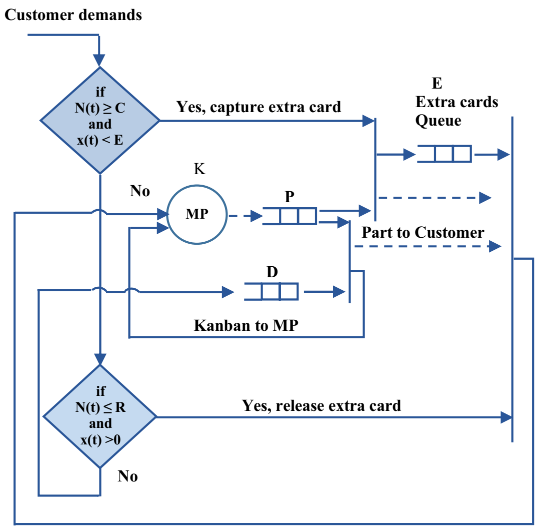

Figure 1. Model for a single-stage adaptive kanban control system [17].

Initially, before a customer demand arrives in the system, queue P has a base stock of K finished pieces, queue E contains all additional cards, queue D is empty, and the manufacturing process MP is inactive. Here, note that C > R and C < K + 1 are necessary consideration for this system to function properly [17]

Design of AKS requires determination of value for number of cards (K), number of extra cards (E), release threshold (R) and capture threshold (C). Several alternative methods are available that employ simple decision rules with thresholds. One such approach, introduced by Tardif and Maaseidvaag [17], focuses on the current inventory and backordered level. To maintain control, an additional card is introduced into the system when demand arises and the inventory level falls below a predetermined release threshold. Conversely, a card is removed from the system when demand occurs and the inventory level exceeds a capture threshold. Tardif and Maaseidvaag devised a local search technique to determine these thresholds, as well as two other parameters crucial for ConWIP system design. However, it is important to note that their method may encounter the challenge of becoming trapped in a local minimum. Shahabudeen and Sivakumar [18] utilized a heuristic-based genetic algorithm to determine the aforementioned parameters, demonstrating the high efficiency of the Genetic Algorithm (GA) through their numerical results. Building upon their work, Sivakumar and Shahabudeen [19] extended their approach to address multi-stage adaptive Kanban systems and found that the Simulated Annealing algorithm (SA) also proved effective in optimizing such systems. In a subsequent study, Sivakumar and Shahabudeen [21] employed a search combining genetic and simulated annealing algorithms to establish threshold values for adaptive Kanban systems, aiming to achieve the same objective. In works presented by Sivakumar and Shahabudeen, the system is modeled in terms of the state (i, x) whose evaluation describes a continuous time Markov chain (CTMC). State (N(t), x(t)) denotes the total number of parts in queue P minus the total number of back orders in queue D by N and the number of extra cards in circulation by x.

Expanding on this objective, Pierreval et al. [6] proposed a simulation-based optimization approach specifically designed for reactive ConWIP systems. In a follow-up study, Belisario and Pierreval [22] employed a simulation-based genetic programming approach to modify the number of cards, thereby reducing WIP while meeting customer demands. Manupati et al. [23] proposed a Support Vector Machine based on simulation approach to tackle a production control problem in a flexible manufacturing cell operating in a dynamic environment, with the aim of maximizing total weighted throughput and minimizing weighted tardiness. Renna [24], propose an adaptive card-based pull control approaches using demand forecasts. The author introduced a methodology involving the development of a fuzzy controller to adjust the number of cards, which has been validated using a simulation module.

In a subsequent work, Elloumi et al. [25,26] propose an intelligent policy based on fuzzy logic for dynamically adjusting the number of Kanban cards in a pull production system operating in a stochastic environment with fluctuations in customer demand. The objective is to maintain a high level of performance measures in terms of production costs and customer satisfaction without any estimation of future demands.

Xanthopoulos et al. [27] proposed a study regarding card-based adaptive control policies that dynamically adjust the number of cards in response to environmental changes, with the aim of reducing backorders and increasing throughput. The authors point out that previous literature has neglected Generalized Kanban and Extended Kanban as adaptive systems, and that no comprehensive experimental study has compared multiple adaptive control policies in a multi-objective context.

Azouz et al [8] presented an analytical study on the concept of nervousness in the context of adaptive control systems by proposing an approach based on multi-objective optimization to reduce the number of Kanban card changes. Then, Azouz and Pierreval [28] exploited an intelligent learning approach based on a neural network to dynamically adjust the number of Kanban cards in real time. The authors exploited a multi-objective optimization technique to determine the thresholds and show its interest compared to other works in the literature by considering nervousness.

Industry 4.0 has revolutionized traditional production systems through digital technologies associated with production systems for efficiency and adaptability. In this context, the concept of electronic Kanban has gained considerable attention as a modern evolution of the traditional Kanban system. In e-Kanban, the Kanban message or order operates in electronic form using information technology tools. The traditional paper-based Kanban system has become obsolete due to various drawbacks. After the introduction of the electronic Kanban, barcodes, QR codes and RFID became common techniques for making information flows visible throughout the production or supply chain to increase the flexibility of systems [29]. Indeed, the adoption of electronic Kanban cards offers several advantages over traditional physical systems, improving efficiency and accuracy in a variety of operational contexts. These benefits include improved data precision, real-time monitoring and improved accessibility, which collectively contribute to better decision-making and operational performance [30,31].

The application of Kanban is not limited to industrial production systems, but also extends to other areas, such as healthcare systems and pharmaceutical stock management. For example, Sriyanto and Ika [32] have demonstrated the effectiveness of e-Kanban in hospital pharmacies to minimize waste, optimize stock levels and simplify management processes. Furthermore, in the automotive industry, Candea and Gabor [33] illustrated that the implementation of the Kanban system as part of a customized approach ensured that the specific needs of companies. Furthermore, Fuentes-del-Burgo et al [34] highlight the success of Kanban in construction, where it is used for supply chain management, Just-In-Time planning and work coordination via the Last Planner System. Their study also highlights the evolution toward e-Kanban systems and their application in Building Information Management environments, as well as the development of new software applications and methods. Zeng et al. [35] demonstrate the use of a building information modeling-enabled e-Kanban system for demand-driven replenishment in construction logistics, resulting in reduced waste, improved scheduling, and site storage optimization. These examples show that Kanban, and e-Kanban in particular, remains a modern and relevant solution in a variety of sectors thanks to its flexibility and alignment with the technologies of the 4th industrial revolution.

These studies demonstrate the polyvalence and adaptability of Kanban, particularly in its electronic form, in addressing management and coordination challenges in a variety of industrial and technological contexts. However, the effectiveness of a Kanban system, and more specifically of an adaptive Kanban, relies heavily on the optimization of its parameters, which enables it to respond to the constant variations of dynamic environments and increasing operational requirements. Indeed, optimizing the parameters of an adaptive Kanban system is crucial, enhancing its flexibility and ability to adapt to the realities of modern industry.

On the other hand, evolutionary algorithms have proven their effectiveness in solving numerous parameter design problems, demonstrating better results than statistical methods [36]. Furthermore, the PSO algorithm is part of evolutionary algorithms. Inspired by the social behavior of insect swarms, it replicates the movement of a set of particles (a swarm) through a multidimensional space to reach the most desirable position. The trajectory of each particle is based not only on its best-achieved position but also on the best position achieved by any particle in the swarm, known to all members of the swarm. Thus, a balance between local and global search is maintained [37]. Unlike conventional optimization methods, it is not a gradient-based method. PSO is considered a robust stochastic optimization algorithm, with less sensitivity to specific algorithm parameters compared to other evolutionary algorithms (e.g. GA). It requires fewer adjustments of specific algorithm parameters than other evolutionary algorithms [38,39].

According to the review presented by Sibalija [40], the PSO has many positive features that have made it very popular in recent years and have expanded its application for various purposes. In addition to its unique property of balancing between local and global search, the simplicity of the PSO algorithm compared to other evolutionary algorithms is particularly appreciated. In the same study, Sibalija also presented the effectiveness of using the multi-objective particle swarm optimization algorithm. The multi-objective approach is commonly employed to solve optimization problems with multiple responses, especially when the number of responses is limited. In multi-objective problems, multiple objective functions may be mutually conflicting, leading to a procedure based on a set of non-dominated solutions, resulting in convergence toward multiple points rather than a single one. In multi-objective PSO the selection of a global leader (gbest) is based on the Pareto dominance principle [40].

The effectiveness of PSO in solving large-scale global problems has been demonstrated in various studies [41–44]. In a study addressing both environmental sustainability and assemblability, Chiu et al. [45] apply the PSO algorithm to calculate an optimal module organization as well as assembly methods and sequences. A case study on an air purifier is presented to illustrate the benefits of the proposed method. As a result, the redesigned product can be more easily maintained during use and recycled at the end of its useful life.

In another research direction, multi-pass turning parameters have been optimized by several authors in the context of designing manufacturing process parameters, as detailed in the review of Sibalija [40]. For instance, Venkata Rao et Pawar [46] used PSO, SA, and artificial bee colony (ABC) for designing key milling parameters (cutting speed, depth of cut, and feed per tooth) considering three constraints with the aim of reducing processing time. The results indicate that PSO and ABC provide a better solution than SA. Furthermore, PSO demonstrated slightly faster convergence than ABC and significantly better convergence than SA.

Li et al. [47] outline the challenges faced by building materials equipment manufacturing enterprises in outsourcing resources and introduces a model and algorithm for supplier selection Traditional methods are insufficient to meet the current requirements for evaluating and selecting outsourced suppliers. Due to traditional methods are insufficient to meet the current requirements for evaluating and selecting outsourced suppliers, a model for outsourcing supplier selection is established, considering factors such as quality, price, delivery time, reliability, and availability of outsourced suppliers.

In this context, this study presents an approach to determine the optimal values of decision variables based on integration of simulation and optimization. This approach utilizes a discrete event simulation software coupled to the metaheuristic Particle Swarm Optimization.

2.2. Use of the simulation

Simulation stands as a widely utilized tool that offers numerous advantages dealing with stochastic and complex dynamic behaviors in a realistic manner, without relying on restrictive assumptions [2,5]. One significant benefit is the ability to incorporate fluctuations in customer demand. Specifically, the average demand rate can be modeled to dynamically evolve over time based on various signals, which can be simulated alongside the stochastic nature of order arrivals. This enables the exploration of scenarios where customer demand is unpredictable, eliminating the need for order arrival information or forecasting [8,25].

Several studies have highlighted the potential of simulation for optimizing and evaluating production systems, particularly in situations where stochastic and complex behaviors need to be taken into account [48]. Jahangirian et al. [49] provide a detailed review of the use of simulation in manufacturing and business sectors, emphasizing that simulation is a useful tool for decision-making, especially in complex or uncertain situations. Iwase and Ohno [50] evaluated the performance of a multistage JIT production system using a simulation model, showing that optimizing production capacity can significantly improve the overall performance of the production system. Additionally, Sagawa and Nagano [51] also used simulation models to model the dynamics of an adaptive multi-product manufacturing system, demonstrating that simulation is an effective tool for managing the stochastic and complex behaviors of the system, as well as for taking into account fluctuations in customer demand.

In an extensive review of recent literature, Ghasemi et al [52] demonstrate the growing interest in applying simulation optimization techniques to production planning particularly in the context of Industry 4.0. This review identifies current gaps, including the need for more integrated frameworks to address uncertainty, optimization and simulation in dynamic environments. Indeed, the focus on Industry 4.0 highlights the potential of Simulation Optimization to provide real-time decision support tools, addressing the planning and control needs of modern production systems.

In the context of integrating optimization with simulation, this study proposes an approach based on optimization via simulation to evaluate the performance of an adaptive Kanban system. Specifically, a discrete-event simulation is employed to reproduce the dynamic behavior of the production system. The PSO algorithm is used to generate potential solutions, which are then evaluated within the simulation framework. This integration enables accurate estimation of the objective function, which aims to reduce production costs while maintaining high levels of service, taking into account the stochastic and dynamic nature of the system. This contribution highlights the effectiveness of the proposed approach in addressing the challenges of parameter optimization for adaptive Kanban systems in complex and variable production environments.

3. Method

3.1. Mathematical formulation of the optimization problem

The objective of the AKS is to minimize the cost associated with the backorder and the inventory. Let $X = [K, E, R, C]$ is the setting vector to be optimized, $X \in S$ with S represents the set of all possible settings. Therefore, the objective function corresponding to the production cost can be expressed as follow:

$$MinZ(X) = \overline{FP} + \overline{WIP} + b.\bar{B} \tag{1}$$

Where

$\overline{FP}$: the average stock of finished products.

$\overline{WIP}$: evaluated as the average difference between the number of products entering and exiting.

$\bar{B}$: represents the average number of demand waiting a finished product.

b : represents the backorder penalty cost.

In order to ensure that the system is able to return to the initial state, following inequality constraints must be satisfied:

$$C \leq K + 1 \tag{2}$$

$$R < C \tag{3}$$

$$K, E, C, R > 0 \tag{4}$$

3.2. Optimization via simulation for design of AKS

The evaluation of the fitness function is a crucial step in the PSO algorithm as it measures the quality of the solution associated with each particle in the search space. To evaluate the fitness function, the PSO algorithm must execute the corresponding Arena model for the decision parameters of the particle being evaluated. The Arena model is a representation of the adaptive Kanban system in terms of processes, resources, workflow, constraints, etc. The model must be compiled to generate the executable file (.p). This file contains the compiled Arena model, ready to be executed by the PSO algorithm. At each iteration, the algorithm evaluates the quality of the solution of each particle using the Rockwell Arena model with the decision parameters of the particle. The PSO algorithm can produce relevant results that reflect the performance of the adaptive Kanban system for that particle in terms of cost. Figure 2 shows the communication flow between the adaptive Kanban system simulated by the discrete event simulation tool Arena and the PSO optimization algorithm developed by the Python programming language.

PSO is a popular metaheuristic algorithm that mimics the behavior of bird swarms and groups of fish. It was first introduced by Kennedy and Eberhart in 1995 for complex, non-linear, multi-dimensional optimization proposals [13]. The algorithm has several advantages, including its simplicity, efficiency, fast convergence and ease of implementation.

3.3. Particle swarm optimization algorithm

PSO algorithm offers several advantages as an optimization technique, particularly in complex problem-solving scenarios like adaptive Kanban systems. We use the PSO for many reasons, which are detailed as follows:

Optimization efficiency: As a stochastic algorithm based on a population approach, PSO is able to efficiently explore large search spaces, which ensures high optimization efficiency. As a result, it is suitable for both single-objective and multi-objective optimization tasks [53].

Versatility and adaptability: The algorithm has been successfully applied in various fields, including health, environmental sciences and industrial applications, demonstrating its flexibility. In addition, different variants of PSO improve its ability to solve complex problems, further extending its applicability [54]. Furthermore, the collective intelligence allows PSO to converge toward optimal solutions faster than many other optimization methods, such as genetic algorithms [55].

Robustness and stability: PSO algorithms demonstrate notable stability and convergence characteristics, which are essential for tackling problems that are poorly defined [56].

Simplicity of implementation: PSO is relatively easy to implement and requires fewer parameters to tune compared to other algorithms, making it accessible for various applications in manufacturing and logistics [57].

Although the PSO algorithm has several advantages, including its simplicity as it requires fewer parameters than many other optimization techniques, its efficiency, rapid convergence and ease of implementation, it is essential to take into account the potential limitations, such as sensitivity to parameter settings and the risk of premature convergence, which can affect its performance in certain scenarios [58].

In PSO, a population of particles (representing candidate solutions) moves through the search space toward the optimal solution based on their own best known position and the global best known position [13]. PSO has been applied successfully to solve various optimization problems, such as function optimization, clustering, and classification. PSO has also been shown to be effective in solving complex optimization problems with high-dimensional search spaces. Figure 3 illustrates the strategy for moving a particle in its environment.

Particle movement is influenced by three main components: the inertia component, the cognitive component and the social component. Each of these components plays a crucial role in determining how particles adjust their position in the search space,

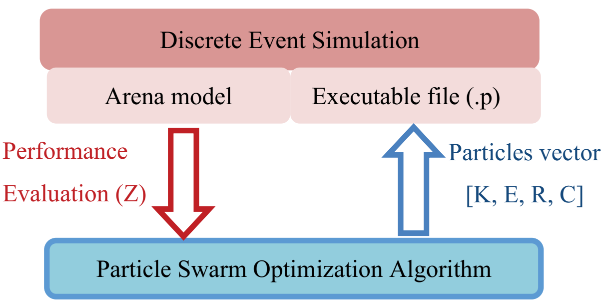

Figure 2. Communication between simulated system and PSO algorithm.

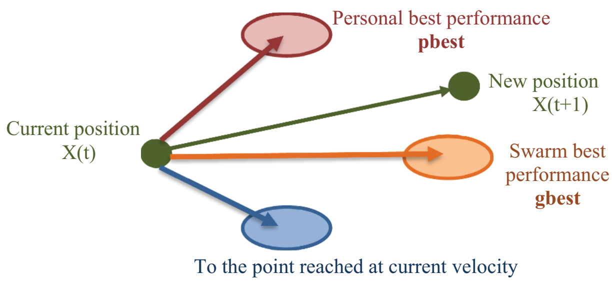

Figure 3. Particle's position displacement.

enabling the PSO algorithm to efficiently search for optimal solutions in complex problem spaces [13]:

- The inertia (physical) component: represents the natural tendency of a particle to continue moving in its current direction as a function of its previous speed, and helps to maintain a balance between exploration and exploitation.
- The cognitive component: reflects the particle's own experience by directing the particle to move toward its personal best position (pbest), which represents the best solution it has found so far. This component encourages individual exploration, allowing particles to refine their search based on their own successes. The cognitive aspect is important to ensure that particles not only rely on the best solutions in the swarm, but also take into account their own discoveries.
- The social component: represents the influence of the swarm as a whole, and encourages the particle to move toward the best global position (gbest), which is the best solution found by all its neighbors. This component encourages collaboration between particles by enabling them to share information about the best areas of the search space. The social aspect enables the swarm to converge toward optimal solutions by taking advantage of collective intelligence.

Each particle of the swarm at time t is characterized by: X (its position in the search space), V (its velocity), pbest (the position of the best solution through which it has passed) and gbest (the position of the best known solution of the whole swarm). The adaptation of the particle between iterations t and t + 1 is done according to the following two equations (5) and (6):

$$V(t+1) = \underbrace{wV(t)}_{inertiacomponent} + \underbrace{c_1 r_1 (pbest(t) - X(t))}_{cognitivecomponent} + \underbrace{c_2 r_2 (gbest(t) - X(t))}_{socialcomponent} \tag{5}$$

$$X(t+1) = X(t) + V(t+1) \tag{6}$$

Where w is the coefficient of inertia which was introduced by Shi and Eberhart [59] to control the influence of the direction of the particle on the future displacement. The purpose of introducing this parameter is to achieve a balance between exploitation (local search) and exploration (global search). Some strategies for adjusting the coefficients include gradually reducing w over iterations to improve convergence [60]. The dynamic inertia coefficient is set according the following equation:

$$w = w_{max} - \frac{w_{max} - w_{min}}{iter} \times t \tag{7}$$

With w_{max} and w_{min} are respectively the initial and the final inertia factor weights, *iter*is the maximum iteration number and t is the current iteration number.

Thanks to its specific balance between local and global search, the PSO algorithm has a lower risk to being trapped in a local solution than the majority of evolutionary algorithms. However, improper configuration of algorithm-specific parameters can lead to premature convergence. According to various authors [37,61], it is crucial to carefully examine these parameters to ensure optimal performance.

c_1 and c_2 are two constants that represent the acceleration coefficients, they can be non-constant in some cases depending on the optimization problem posed [13]. As indicated in (5), constants c_1 and c_2 pulls each particle toward *pbest* positions (cognitive component of velocity) and *gbest* positions (social component of velocity). r_1 and r_2 are two random numbers generated in the interval [0,1].

The main steps of the PSO algorithm are as follows [13]:

- 1) Initialization: A number of particles are initialized with random positions in the search space, and each particle is assigned an initial velocity.
- 2) Evaluation: The quality of each solution represented by the particles' positions is evaluated using the cost function.
- 3) Update of personal best solution (pbest): If a particle finds a more efficient solution than its previous personal best solution, this new solution is recorded as the new personal best solution.
- 4) Update of global best solution (gbest): The most efficient personal solution among all particles is recorded as the global best solution.
- 5) Update of velocity: The velocity of each particle is updated based on its own position, personal best solution, and global best solution using (5).
- 6) Update of position: The position of each particle is updated based on its velocity using (6)
- 7) Repeat steps 2–6 until the stopping criterion is reached, typically a maximum number of iterations.
- 8) Return the recorded global solution as the optimal solution.

3.4. Determination of the control parameters

In this work, each particle of the swarm is represented by a vector of decision variables $X = [K, E, R, C]$ representing number of kanban cards, number of extra cards, release threshold, and capture threshold, respectively.

3.4.1. PSO initialization

Initialize randomly the particle population array. Initial population is generated randomly subjected to constraints (2), (3) and (4). To do this, the decision variables are initialized randomly as expressed by (8), knowing that K_{max} and E_{max} designate respectively the maximum number of fixed cards and that of additional cards.

$$\begin{gathered} \textit{Select random value of } K \textit{ in the interval } [1, K_{max}] \\ \textit{Select random value of } E \textit{ in the interval } [1, E_{max}] \\ \textit{Select random value of } C \textit{ in the interval } [2, K+1] \\ \textit{Select random value of } R \textit{ in the interval } [1, C-1] \end{gathered} \tag{8}$$

Let us note that the population size of PSO algorithm should be large enough to adequately cover the solution space without incurring excessive computational effort.

3.4.2. Correction and updating of velocity and position

The position and velocity are updated according (5) and (6). Furthermore, when the particles in the PSO algorithm must respect constraints, such as a domain' membership, it is important to adapt the model to consider them. A constraint checking strategy has been implemented to handle the constraints in the PSO algorithm. After having updated the position of the particle, we check if it respects the constraints (2), (3) and (4). If this is not the case, the position of the particle is readjusted or repaired by selecting a random value that respects the constraints as expressed by (8).

4. Results

4.1. Description of the studied system

In this section, we introduce the single-stage adaptive kanban model used in this study. For comparative purposes, the production system considered in this study is taken from Tardif and Maaseidvaag [17], which is considered a widely used example in the literature. The manufacturing process considered consists of a single stage of workstation with 10 parallel servers where processing times are exponentially distributed with a mean of 6 (or μ = 1/6). The initial state before receiving any demand is specified as follows: there exists an initial stock of K finished products prepared for customer delivery, no demands are backordered, and the manufacturing process is idle.

The simulation models were developed using Arena simulation software (version 14) and the PSO algorithm was developed using the Python programming language under Visual Studio Code (version 1.87). All experiments were carried out on a PC equipped with an Intel Core (TM) i5–10505 processor at 3.20 GHz, 8.00 GB RAM and running a Windows 10 Professional 64-bit operating system.

In this study, we will consider four cases that describe different patterns for the customer demands. As follow, we summarize the numerical experiments' parameters of each case.

- Case 1: The demand follows the Poisson process with rate $\lambda_d = 1/7$ (demand mean = 7), the backorder penalty cost "b" is 500 [17,19].
- Case 2: The demand follows the Poisson process with rate$\lambda_d = 0.2$ (demand mean = 5), the backorder penalty cost "b" is 1000 [17,19].
- Case 3: The arrival of demands is a Poisson process follows a triangular pattern that linearly increases from 3 to 7 (or λ_d from 1/3 to 1/7) over 75 time units. It then decreases linearly from 7 to 3 over the next 75 time units, and this pattern repeats throughout the simulation run as shown in Figure 4(b). the backorder penalty cost "b" is 1000 [17,62].
- Case 4 : Demand arrivals follow a cyclic Poisson process with varying means. Specifically, the demand mean for the first 25 time units is 3, for the next 25 time units it is 5, and for the last 25 time units it is 7 as shown in Figure 4(a). This pattern repeats throughout the simulation run [22,62].

4.2. Computational results

To evaluate the validity of our approach, we used the same simulation parameters as those adopted in the works considered as study for comparison. The

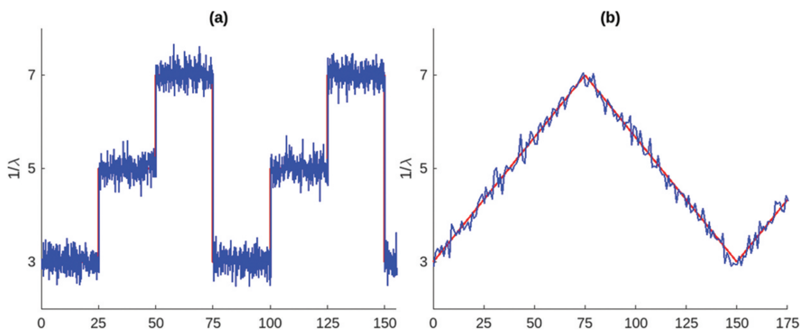

Figure 4. Evolution of the distribution of the average customer demand.
(a) case 4: cyclical pattern (b) case 3: triangular pattern.

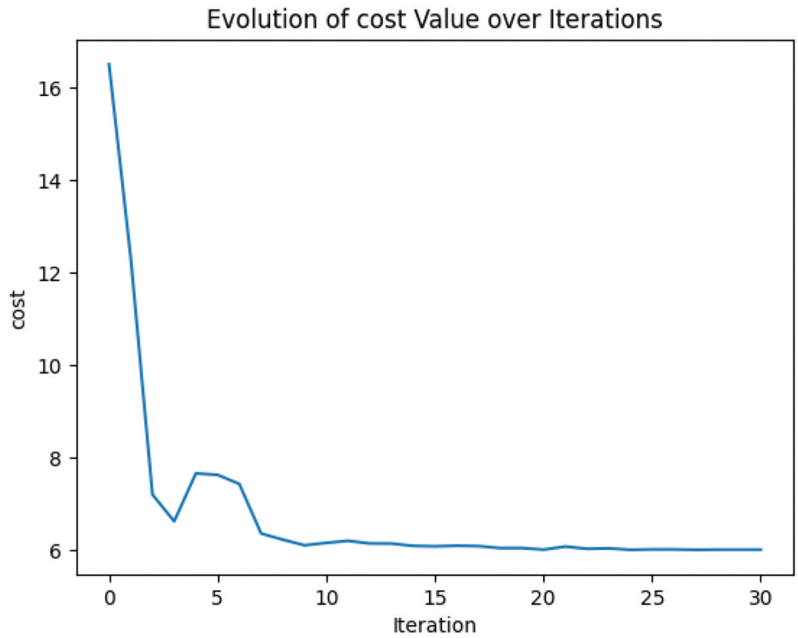

Figure 5. Evolution of the objective function during the optimization via simulation.

simulation parameters were set as follows: replication length is equal to 1,000,000 time units, number of replications is 30 and warm-up period is equal to 1000.

Figure 5 shows the evolution of the objective to be minimized during the iterations for the third case where the customer demand follows a triangular pattern.

In our proposed PSO algorithm, the population size and number of generations are parameters related to the convergence of the algorithm. In the proposed approach, a population size of 20 and a number of generations equal to 30 were used. The individual learning factor c_1 and social learning factor c_2 are both fixed to 0.5. r_1 and r_2 are two random numbers generated in the interval [0,1]. The parameters of the dynamic inertia coefficient w are respectively $w_{min} = 0.5$ and$w_{max} = 1$. Finally, the limits of the design variables of the optimization problem K_{max} and E_{max} are fixed respectively to 15 and 10.

The Table 1. illustrates the optimized values of the decision vector variables $X^* = [K, E, R, C]^*$ and the minimum cost value Z obtained for the different optimization algorithms (LSM, GA, SA, and PSO), as well as for the traditional Kanban system (TKS).

5. Discussions

As it is recommended to compare the performance of PSO with other optimization methods suitable for the specific problem to assess its relative effectiveness, we will conduct a comparison in this section. We compare the results obtained by the PSO algorithm for the AKS with those obtained using the Local Search Method (LSM) proposed by Tardif and Maaseidvaag [17], as well as those obtained using GA and SA proposed by Shahabudeen and Sivakumar [19]. These results are also compared to those obtained using the traditional Kanban system [17]. Overall, these results confirm that the integration of PSO algorithm for the design of an adaptive Kanban system is not only more efficient than other metaheuristic methods such as GA, SA and LSM, but also offers more significant improvements compared to traditional Kanban systems in terms of production cost minimization.

However, there are also limitations to consider. PSO can be sensitive to parameter choices, and poor configuration can lead to premature convergence. Additionally, PSO's performance can vary depending on the nature of the problem, and it is essential to understand its characteristics for effective application. Therefore, it is recommended to carefully adjust the algorithm parameters based on the specific nature of the problem to be solved. In our approach, several preliminary tests and iterative adjustments of parameters were conducted to achieve good results.

The results presented in Table 1 affirm the advantages offered by PSO. This algorithm demonstrates superior performance in minimizing production costs compared to other metaheuristic algorithms. While other evolutionary algorithms involve competition among individuals in a population, PSO also incorporates a cooperative strategy. This unique property of balancing local and global search distinguishes PSO from other algorithms in overcoming the issue of premature convergence associated with all metaheuristics.

The proposed approach of optimizing through discrete event simulation using the PSO algorithm has demonstrated its efficacy. Table 1 illustrates that the

Table 1 Comparison of simulation results.

	TKS [17]		*LSM* [17]		*GA* [19]		*SA* [19]		*Proposed approach PSO*	
cases	Z	K	Z	X*	Z	X*	Z	X*	Z	X*
1	5.2	5	5.1	5, 6, 1, 2	4.7	3, 12, 3, 4	4.53	4, 6, 2, 3	4.41	3,2,3,4
2	6.39	6	5.61	4, 6, 3, 4	5.61	4, 6, 3, 4	5.61	4, 6, 3, 4	4.89	4,3,3,4
3	7.18	6	-	-	-	-	-	-	5,99	5, 5,3,4
4	7.35	7	6.25	4, 6, 3, 4	-	-	-	-	6.11	5, 5,3,4

Table 2 Percentage improvement achieved by PSO algorithm.

cases	*TKS* [17]	*LSM* [17]	*GA* [19]	*SA* [19]
1	15.19%	13.53%	6.17%	2.64%
2	23.47%	12.83%	12.83%	12.83%
3	16.57%	-	-	-
4	16.87%	2.24%	-	-

decision vector parameters for the design of an adaptive Kanban system obtained by PSO result in better outcomes in terms of minimizing production costs.

Table 2 illustrates the percentage improvement achieved by our approach compared to other approaches in terms of the objective function.

For pull systems, taking into account the stochastic fluctuation of customer demand is crucial. The design of an adaptive kanban system with optimal decision parameters enables us to overcome the effects of these fluctuations. The proposed method has demonstrated convergence even in cases where the average demand is fluctuating, as shown in cases 3 and 4. This demonstrates that PSO is regarded as a robust stochastic optimization algorithm, in terms of being less sensitive to specific algorithm parameters compared to other evolutionary algorithms (GA and SA).

6. Concluding remarks

In conclusion, the proposed PSO-based approach provides a promising solution for the design and optimization of an adaptive Kanban system. This dynamic adjustment approach allows the system to consider stochastic and time-changing demand, while simultaneously optimizing key performance metrics such as inventory cost. The simulation coupled to optimization framework ensures that the system is optimized for realistic scenarios and allows for evaluation of different scenarios before implementation. The results of the case study demonstrate the effectiveness of the proposed approach in improving system performance and highlight its potential for application in a range of manufacturing environments. More precisely, our results indicate that the proposed approach has provided an improvement of up to 23% compared to classical approaches and up to 12% compared to other metaheuristics such as GA and SA. Overall, the PSO-based approach represents a valuable contribution to the field of adaptive Kanban systems and provides a useful tool for practitioners seeking to enhance the efficiency and responsiveness of their production systems.

However, this study is not without its limitations. First, the model developed relies on specific assumptions regarding demand patterns and system behavior, which may not fully capture the complexities of all real-world environments. Future research could explore a wider range of demand scenarios, including extreme fluctuations and long-term trends, to validate the robustness of the proposed approach.

Furthermore, this study focused primarily on a single production environment; thus, its findings may not be generalizable across different industries or production settings. Expanding the scope of future studies to include diverse manufacturing contexts would provide a more comprehensive understanding of how adaptive Kanban systems can be optimized.

Despite the promising results of this study, several perspectives for future research are opened. First, integrating advanced machine learning techniques, such as reinforcement learning or deep learning, could enhance the decision-making process and improve the system's ability to eliminate the effect of unpredictable demand fluctuations on production costs. Secondly, increased product diversification makes planning, scheduling and balancing production flows even more complex. The ability to manage these challenges effectively has become crucial for companies operating in multi-product environments. Developing adaptive and intelligent management strategies is becoming essential to meet the challenges of a multi-product and uncertain environment. Thirdly, a multi-objective optimization approach using PSO could be developed to address not only production cost minimization and customer satisfaction, but also the problem of over-correction in terms of the frequency of Kanban card adjustments. This extension would allow the system to achieve a more stable and efficient production flow while maintaining its capacity to be adapted. These lines of research offer great potential for improving the adaptability and robustness of Kanban-based production systems.

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No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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